

Exercise 1.6.2 Find the line element ds^2 for the following surfaces: (a) Sphere of radius a , (b) Cylinder of radius a , (c) Hyperbolic paraboloid of Example 1.6.1

Solution.

One way to find the line element is to directly use the line element formula for a surface:

$$ds^2 = g_{AB} du^A du^B \text{ for } A, B = 1, 2$$

u^A is known in all three cases:

(a) $u^1 = \theta$ and $u^2 = \phi$

(b) $u^1 = \theta$ and $u^2 = z$

(c) $u^1 = u$ and $u^2 = v$

If we don't already have the metric tensor components, we get it from

$$g_{AB} = \mathbf{e}_A \cdot \mathbf{e}_B$$

by computing

$$\mathbf{e}_A = \frac{\partial \mathbf{r}}{\partial u^A} = \frac{\partial x}{\partial u^A} \mathbf{i} + \frac{\partial y}{\partial u^A} \mathbf{j} + \frac{\partial z}{\partial u^A} \mathbf{k}$$

where

$$\mathbf{r} = x \mathbf{i} + y \mathbf{j} + z \mathbf{k}.$$

Here we use a direct approach, starting with $\mathbf{r}$, computing $d\mathbf{r}$, and then

$$ds^2 = d\mathbf{r} \cdot d\mathbf{r}$$

(a) Sphere of radius a

Fixing $r = a$, then from Example 1.1.4,

$$\mathbf{r} = a \sin \theta \cos \phi \mathbf{i} + a \sin \theta \sin \phi \mathbf{j} + a \cos \theta \mathbf{k}$$

$$\begin{aligned} d\mathbf{r} = & a (\cos \theta \cos \phi d\theta - \sin \theta \sin \phi d\phi) \mathbf{i} \\ & + a (\cos \theta \sin \phi d\theta + \sin \theta \cos \phi d\phi) \mathbf{j} \\ & - a \sin \theta d\theta \mathbf{k} \end{aligned}$$

$$\begin{aligned} ds^2 = d\mathbf{r} \cdot d\mathbf{r} \\ = & a^2 [(\cos^2 \theta \cos^2 \phi + \cos^2 \theta \sin^2 \phi + \sin^2 \theta) d\theta^2 \\ & + (\sin^2 \theta \sin^2 \phi + \sin^2 \theta \cos^2 \phi) d\phi^2] \\ = & a^2 (d\theta^2 + \sin^2 \theta d\phi^2) \quad \checkmark \end{aligned}$$

(b) Cylinder of radius a

Fixing $r = a$, then from Exercise 1.1.1,

$$\mathbf{r} = a \cos \phi \mathbf{i} + a \sin \phi \mathbf{j} + z \mathbf{k},$$

$$d\mathbf{r} = -a \sin \phi d\phi \mathbf{i} + a \cos \phi d\phi \mathbf{j} + dz \mathbf{k}$$

$$ds^2 = d\mathbf{r} \cdot d\mathbf{r}$$

$$= a^2 (\sin^2 \phi + \cos^2 \phi) d\phi^2 + dz^2 = a^2 d\phi^2 + dz^2 \quad \checkmark$$

(C) Hyperbolic paraboloid

$$\mathbf{r} = (u + v) \mathbf{i} + (u - v) \mathbf{j} + 2uv \mathbf{k}$$

$$d\mathbf{r} = (du + dv) \mathbf{i} + (du - dv) \mathbf{j} + 2(v du + u dv) \mathbf{k}$$

$$ds^2 = d\mathbf{r} \cdot d\mathbf{r}$$

$$= 2(1 + 2v^2) du^2 + 2(1 + 2u^2) dv^2 + 8uv du dv \quad \checkmark$$